# Srinivasan Balasubramanian

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## Summary

Machine Learning Engineer (B.E. Information Science, 2025) who builds and deploys end-to-end ML systems, not notebooks. Shipped a hybrid RAG system over 2,000+ RBI regulations, a credit-risk MLOps pipeline (MLflow, DVC, drift-triggered retraining) with a live FastAPI service, and a LoRA fine-tuning pipeline for structured extraction. Every project below has public code, tests and CI.

## Technical Skills

**Languages:** Python, SQL, C++, Java, Kotlin
**ML & Deep Learning:** Scikit-Learn, XGBoost, LightGBM, CatBoost, TensorFlow/Keras, OpenCV, Optuna, SHAP
**LLMs & GenAI:** RAG, Hugging Face Transformers, TRL (LoRA fine-tuning), Qdrant, BM25, Cross-Encoder Reranking, Ollama, llama.cpp/GGUF Quantization
**MLOps & Deployment:** MLflow, DVC, Prefect, Evidently, Docker, FastAPI, Flask, GitHub Actions (CI/CD), Pytest, Linux
**Data & Analytics:** pandas, NumPy, DuckDB, PostgreSQL, MySQL, MongoDB, Streamlit, A/B Testing, Statistical Inference

## Experience

**AIMPLIFY** **Jul 2024 – Sep 2024**
*Software Intern (AI/ML)* *Bangalore, India*

- Built Python data-processing workflows to clean and structure unstructured text corpora for LLM and ML model evaluation.
- Designed and benchmarked baseline LLM prompts across test suites; wrote Scikit-Learn cross-validation and error-analysis scripts tracking Precision, Recall and F1.
- Integrated ML inference outputs into internal application prototypes through Flask REST APIs, working with the software team.

## Projects

**RBI Regulations Q&A – Hybrid RAG with Citations** | *Python, Qdrant, FastAPI, Docker* **Code** | **Live Demo** | 2026

- Built a RAG system over 2,000+ Reserve Bank of India notifications: BM25 + dense retrieval (BGE embeddings in Qdrant) merged with Reciprocal Rank Fusion, then cross-encoder reranking, with numbered source citations in every answer.
- Added a calibrated confidence gate (threshold tuned for refusal F0.5 on a document-level split) so the system answers "I don't know" instead of hallucinating on out-of-scope questions.
- Built an evaluation harness (Recall@k, MRR, nDCG, NLI faithfulness, citation accuracy, LLM-as-judge, bootstrap CIs) with a 36-config retrieval ablation automated as a daily GitHub Actions workflow.

**Credit Default Prediction – End-to-End MLOps** | *LightGBM, MLflow, DVC, FastAPI* **Code** | **API** | **Site** | 2026

- Trained, tuned (Optuna) and calibrated LogReg, XGBoost and LightGBM on 30,000 credit-card customers; champion LightGBM scored ROC-AUC 0.778 and PR-AUC 0.536 on a 6,000-customer holdout.
- Replaced the default 0.5 cutoff with a cost-optimal decision threshold (5:1 miss-to-false-alarm cost), cutting expected cost per applicant by 26% and raising default recall from 35% to 77%.
- Shipped the full lifecycle: Pandera data validation, DVC versioning, MLflow registry with an automated promotion gate, SHAP explanations per prediction, Evidently drift detection that triggers retraining, Docker deploy to Render via GitHub Actions CI/CD.

**Job-Post to JSON Extractor – LoRA Fine-Tuning** | *Qwen2.5, HF TRL, llama.cpp* **Code** | **Site** | 2026

- Building a LoRA fine-tune of Qwen2.5-0.5B-Instruct (TRL SFT, completion-only loss) to convert noisy Indian job posts (WhatsApp forwards, Hinglish) into schema-validated JSON across 11 fields, returning null instead of guessing missing values.
- Generated a 3,000-example synthetic dataset with MinHash de-duplication and template-disjoint splits to prevent leakage; scored with field-level exact match, micro/macro F1 and hallucination rate with bootstrap CIs (regex baseline: 93.1 micro-F1).
- Built the 4-bit GGUF quantization and in-browser serving path (llama.cpp WebAssembly) with JSON-grammar constrained decoding.

**E-Commerce Analytics, 99K Orders (Olist)** | *SQL, DuckDB, PostgreSQL, Streamlit* **Code** | **Dashboard** | 2026

- Wrote 16 analytical SQL queries (CTEs, window functions) over 99,441 orders, cross-checked cell by cell in CI on DuckDB and PostgreSQL 16.
- Showed late deliveries drop average review score from 4.29 to 2.27 stars (bootstrap CI, ordinal regression) and found an RFM-based segment holding 12.4% of customers but 38.6% of revenue.
- Ran a simulated A/B test of free shipping (power analysis, SRM check, Bayesian posterior): uplift was significant, but subsidy cost R$9.60 per order for R$1.44 extra revenue, so recommended against launch.

**Multi-Modal Deepfake Detection** | *TensorFlow, OpenCV, Flask, React, Docker* **Code** | **Demo** | 2025

- Built a video, image and audio deepfake detector: Xception and InceptionV3 face models (two-stage transfer learning) with attention pooling over frames, plus a log-mel spectrogram CNN for audio, combined by late fusion.
- Added Grad-CAM explanations; served through a Flask REST API (async jobs) and React dashboard, containerized with Docker.

**Other projects:** Abalone Age Regression (blend/stacking ensembles, RMSLE 41.6% below a mean baseline, FastAPI, in-browser XGBoost); Android e-Challan App (Kotlin, Jetpack Compose, MVVM, Firebase, offline sync, CI-built APK); Redis Chat Service (Spring Boot, Redis, Docker).

## Education

**BMS College of Engineering** **Dec 2021 – Jun 2025**
*Bachelor of Engineering, Information Science and Engineering – CGPA 8.5/10* *Bangalore, India*

## Certifications

Fundamentals of Red Hat Enterprise Linux – Red Hat, Coursera (verify) | Data Analysis with R – Johns Hopkins, Coursera (verify)